Karan Bhoriya

AI Engineer

8/257, Mehram Nagar Delhi Cantt-10, New Delhi, Delhi | +91 9899213147 | **Karanpr1806@gmail.com** | **https://www.linkedin.com/in/karan-bhoriya-b5a3382b7/** | **https://github.com/Karanpr-18** | **https://karanpr-18.github.io/Karan-Portfolio/**

OBJECTIVE

A research driven data scientist who likes to build new innovative models and analyze insights from data. Leveraging my strengths to achieve company's strategic goals.

EDUCATION

Sushant University, Gurugram **2023-Present**

B. Tech CSE (specialization in AIML)

INTERNSHIP

Prompt engineer and Data analyst **April 25–August 25**

Humana People to People India, Vasant Kunj

- Engineered Python validation pipelines and automated schema checks to eliminate manual entry errors, ensuring 100% data integrity and slashing QA time from 4 days to 2 hours.
- Applied LLM prompt engineering to accelerate Python development and automate unstructured data cleaning, alongside developing a geographic code matching utility that reduced reconciliation time by 87%.
- Developed dynamic Power BI dashboards to translate complex data pipelines into actionable, data-driven insights for key stakeholders.

PROJECTS

TalentAI: AI-Powered Recruitment & HR Ecosystem **February 2026**

- Engineered an end-to-end AI recruitment platform using Python and Flask, integrating Google Gemini Pro and Groq Llama 3.1 for NLP-based PDF resume parsing.
- Developed a weighted candidate ranking engine (Technical 50%, Experience 30%, Academics 20%) alongside core HR management CRUD workflows.
- Leveraged Pandas to generate analytical reports and dynamic leaderboards, enabling data-driven candidate screening and hiring decisions.

Job Matcher **January 2026**

- Developed an NLP-powered Streamlit utility to quantify candidate-job compatibility, optimizing the matching engine for sub-second real-time inference.
- Implemented TF-IDF vectorization and Cosine Similarity algorithms to calculate semantic "Match Percentage" scores based on skill density.
- Engineered a robust preprocessing pipeline (lemmatization, entity extraction, stop-word removal) to efficiently handle and parse noisy PDF resume data.

Next Word Prediction **January 2026**

- Engineered a next-word prediction model from scratch using PyTorch, designing and training a feedforward neural network for sequential language learning.
- Constructed a complete text preprocessing pipeline, including custom tokenization and vocabulary generation.
- Evaluated model predictions to validate contextual understanding and comprehensively documented the implementation for educational reproducibility.

Heart Attack Classifier **July 2026**

- Developed a Random Forest classifier to predict heart disease risk using 14 clinical features, applying robust preprocessing and encoding techniques.
- Evaluated model performance and interpretability using confusion matrices, ensuring reliable predictive capabilities.
- Serialized trained models using joblib for reusable inference and published the complete, reproducible notebook workflow with structured documentation.

Stock Price Prediction **July 2025**

- Developed a predictive Linear Regression model for NSE stocks using the yfinance API, engineering lag-based features to adapt time-series data for supervised learning.
- Achieved a highly accurate 99.48% R^2 score and an RMSE of ₹1,572.19, strictly applying time-series train-test splitting to prevent data leakage.
- Visualized actual versus predicted stock prices using Matplotlib to validate model performance and financial trends.

Mail Classifier **August 2025**

- Built an email spam classification system by training a Naive Bayes model on labeled datasets, utilizing TF-IDF vectorization for text feature extraction.
- Developed and deployed a Streamlit web interface integrated with serialized models for real-time spam/ham prediction.
- Documented the end-to-end ML workflow in Jupyter Notebooks—including EDA and preprocessing—ensuring strict reproducibility and dependency management.

AI News Analyzer **September 2025**

- Developed a multi-model NLP News Intelligence Toolkit to classify Fake News, Hate Speech, and News Categories via a public Streamlit application.
- Built an end-to-end ML pipeline using TF-IDF feature engineering and linear classifiers, efficiently processing datasets ranging from 21K to 217K samples.
- Achieved robust model accuracies of 89% (Fake News), 80% (Hate Speech), and 89% (News Category) for real-time news analysis.

Excel Cleaner **August 2025**

- Developed a full-stack Flask web application to automate Excel dataset cleaning and preprocessing, significantly reducing manual data preparation efforts.
- Implemented user-friendly file upload pipelines with modular backend components for secure file handling, data validation, and automated transformations.
- Structured the application architecture using templates, static assets, and routing layers to ensure a highly maintainable and scalable Python web solution.

Used Car Price Analysis **June 2025**

- Delivered an end-to-end used car price analysis pipeline, encompassing EDA, SQL-based data cleaning, and predictive modeling.

- Implemented a dynamic price estimation algorithm utilizing progressive constraint relaxation, converting processed data into lightweight JSONs for fast browser-side inference.
- Built and deployed a responsive, serverless frontend optimized for GitHub Pages hosting, adhering to a professional modular directory structure.

PUBLICATIONS

A study on Human-Centered Emotion Classification based on Classical Machine Learning Models Capture in GoEmotions (Ongoing) **January 2026**

- Conducted a research study evaluating classical ML models (Linear Regression, LinearSVC, Random Forest) on the 58K-sample GoEmotions dataset for imbalanced multi-emotion classification.
- Designed a comprehensive NLP preprocessing pipeline—incorporating redundancy removal, Ekman taxonomy mapping, lemmatization, and TF-IDF/CountVectorizer feature engineering.
- Achieved benchmark metrics of ~53% accuracy and 43% Macro F1-Score, demonstrating the viability of computationally efficient explainable models; manuscript currently in preparation for academic submission.

CERTIFICATIONS

- Data Science with Python for beginners **November 2025**
- SQL Techniques for Data Science **February 2025**

EXTRACURICULARS

Community Service Director **August 2024 – July 2025**
Rotaract Club of Sushant University

Member **August 2023 – July 2024**
Rotaract Club of Sushant University

SKILLS

- **Technical skills**
  - **Programming Languages:** Python, Java, C, C++, SQL, HTML, CSS, JS
  - **Computer software/ frameworks:** Keras, Git, Power BI, Canvas, Linux, Jupyter, TensorFlow, PyTorch, Streamlit, SpaCy, Scikit-Learn, Pandas, NumPy, Matplotlib
- **Soft skills** : Analytical thinking, Problem Solving, Communication, Teamwork, Time management, Critical Thinking. Accountability

LANGUAGES KNOWN

- Hindi
- English